North South University

Department of Electrical & Computer Engineering

Project Proposal

Project Name: PerformTrack (Employee Performance Tracking System)

Github Repository Link: https://github.com/tjnblve-source/CSE327_Project-Employee_Performance_Tracking_System.git

Group Members:

Name	ID
Umme Habiba Haque	2211467042
Farhana Afrin Nahin	2221056042
Ilma Ibrahim	2321019042
Abu Bakar Siddik	2421953042

Submitted to:

Faculty Name: AKM Iqtidar Newaz

Faculty Initial: IqN

Course Code: CSE327.7

Course Title: Software Engineering

Introduction

PerformTrack is a business-focused platform designed to help organizations manage employee performance evaluation and promotion decisions efficiently. In many companies, employee performance tracking and appraisal processes are handled manually or through scattered systems, which can lead to errors, bias, and inefficiency. PerformTrack aims to solve these problems by providing a centralized system where employee data, performance records, attendance, and appraisal history can be stored and analyzed.

Objective of the System

The main objective of PerformTrack is to simplify employee performance evaluation and ensure fair promotion decisions using data-driven analysis. The system allows managers and HR departments to track employee activities, evaluate performance, and monitor progress over time.

System Overview

In the PerformTrack system, employees are registered with important details such as their department, role, and identification information. Managers can periodically evaluate employee performance by assigning ratings, providing feedback, and recording achievements.

The system also tracks employee attendance and maintains appraisal history. Using these data points, PerformTrack analyzes performance trends and identifies employees who qualify for promotions or rewards.

Employees can log into the system to view their performance ratings, feedback, and improvement areas. Meanwhile, HR and management can generate reports to evaluate overall workforce productivity and make strategic decisions.

Benefits of the System

- Reduces manual performance tracking
- Ensures fair and transparent promotion decisions
- Helps employees understand their growth path
- Improves organizational productivity
- Provides data-driven insights for HR management

End Users of the System

1. Employees

Employees can:

- View their performance ratings
- Read feedback from managers
- Track their progress and achievements
- Check promotion eligibility

2. Managers

Managers can:

- Evaluate employee performance
- Provide feedback and ratings
- Record employee achievements
- Review attendance and appraisal records

3. HR Administrators

HR personnel can:

- Manage employee records
- Monitor performance reports
- Approve promotions
- Generate workforce analytics

Functional Requirements

1. The system shall allow HR to register employees.
2. The system shall store employees' personal and job information.
3. Managers shall be able to record performance ratings.
4. Managers shall be able to provide feedback and comments.
5. The system shall track employee attendance.
6. The system shall maintain appraisal history.
7. The system shall calculate performance scores.

8. The system shall identify employees eligible for promotions.
9. Employees shall be able to view their performance reports.
10. HR shall be able to generate performance analytics reports.
11. The system shall support secure login for different users.

<u>Non-Functional Requirements</u>

Security

- Only authorized users can access the system.
- Employee information must be protected.

Performance

- The system should respond quickly to user requests.
- The system should handle multiple users simultaneously.

Usability

- The interface should be easy to use.
- Users should be able to navigate the system easily.

Reliability

- The system should store data without loss.
- The system should operate with minimal downtime.

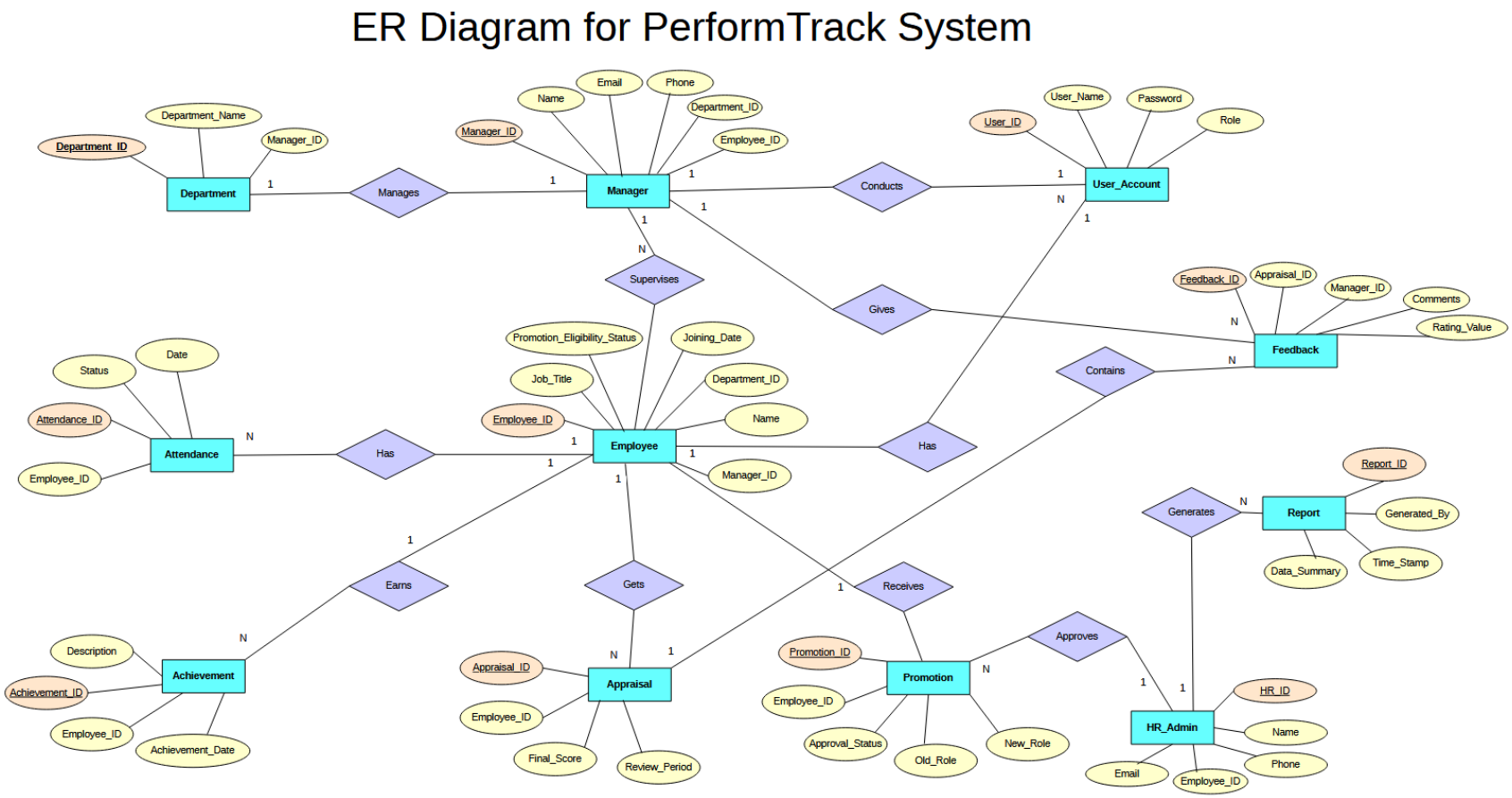
Scalability

- The system should support increasing numbers of employees.

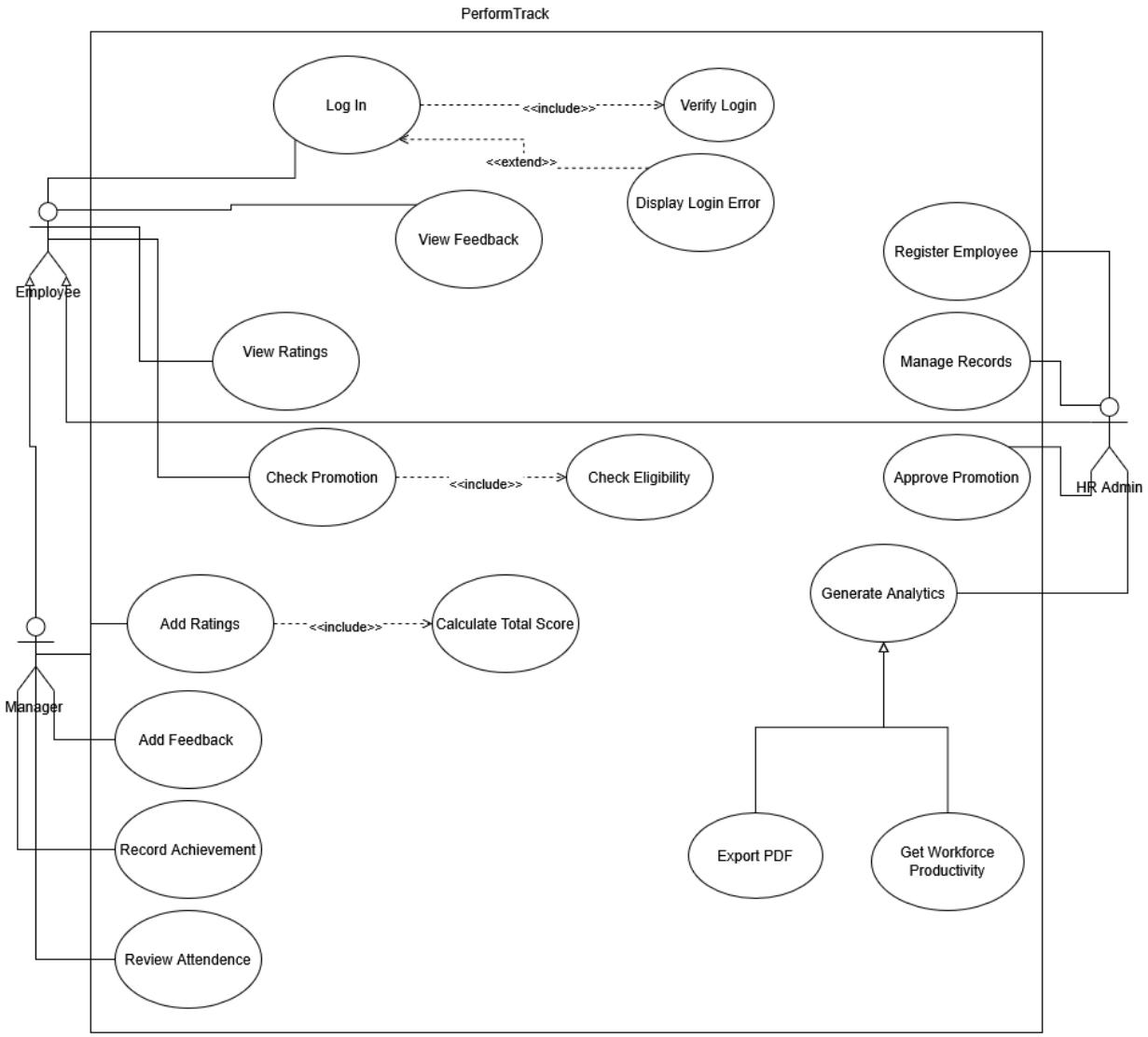
Maintainability

- The system should be easy to update and maintain.

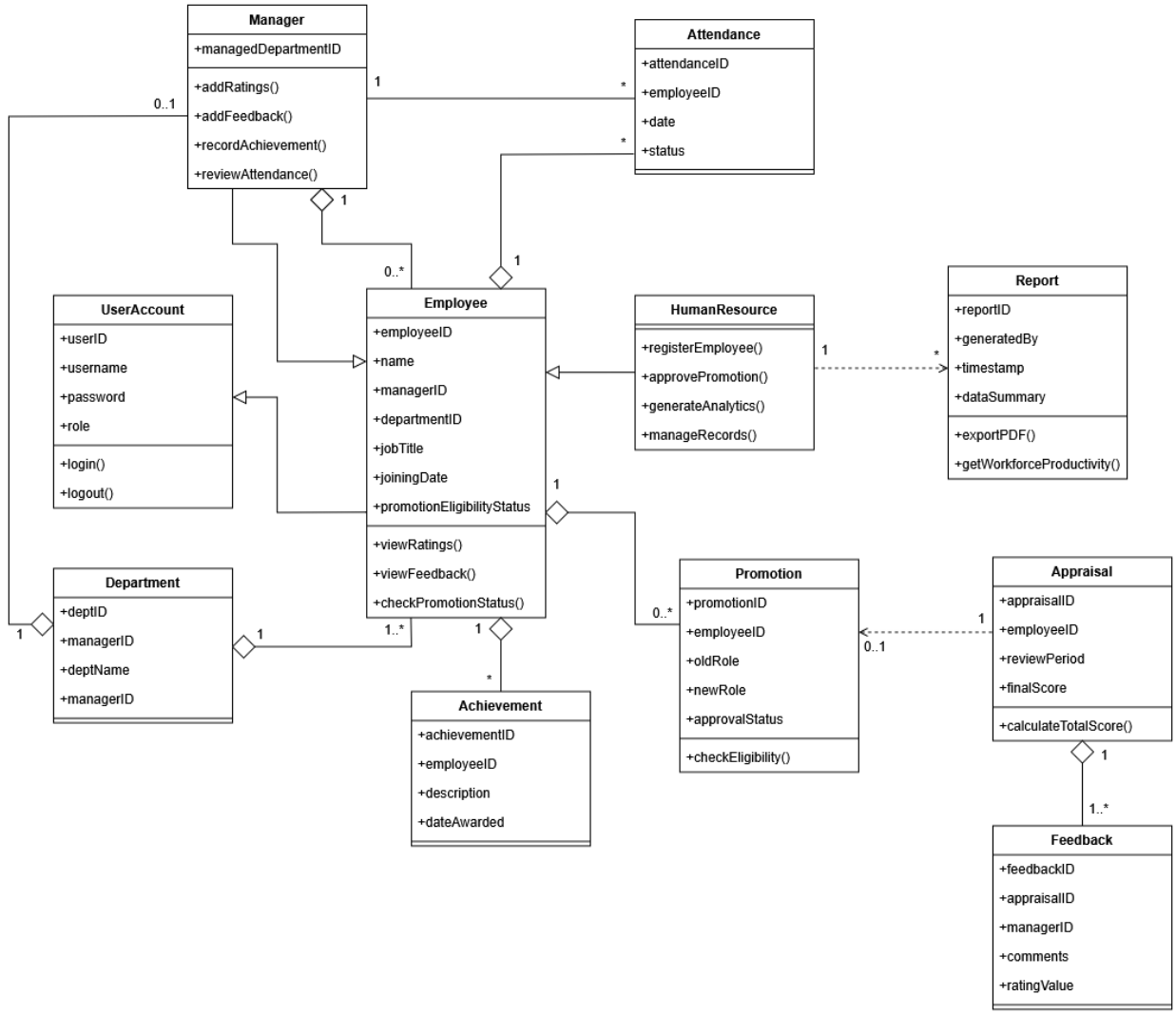
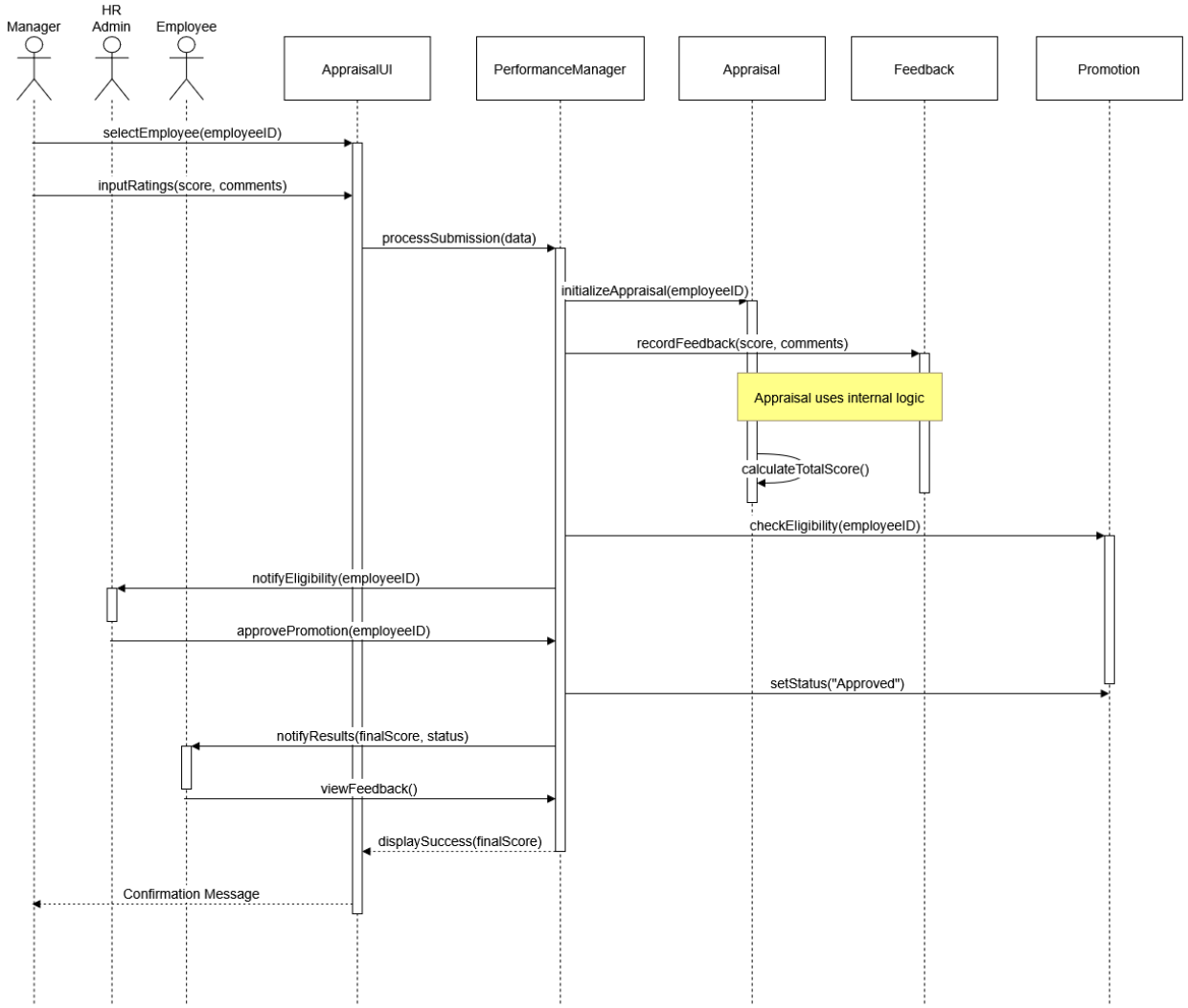
ER Diagram


ER Diagram for PerformTrack System
Department
Department_ID
Department_Name
Manager_ID
Manages
Manager
Manager_ID
Name
Email
Phone
Department_ID
Employee_ID
Conducts
User_Account
User_ID
User_Name
Password
Role
Supervises
Gives
Feedback
Feedback_ID
Appraisal_ID
Manager_ID
Comments
Rating_Value
Contains
Attendance
Attendance_ID
Status
Date
Employee_ID
Has
Employee
Employee_ID
Job_Title
Promotion_Eligibility_Status
Joining_Date
Department_ID
Name
Manager_ID
Has
Report
Report_ID
Generated_By
Time_Stamp
Data_Summary
Generates
Earns
Gets
Receives
Achievement
Achievement_ID
Description
Employee_ID
Achievement_Date
Appraisal
Appraisal_ID
Employee_ID
Final_Score
Review_Period
Promotion
Promotion_ID
Employee_ID
Approval_Status
Old_Role
New_Role
Approves
HR_Admin
HR_ID
Name
Phone
Email
Employee_ID

Use-case Diagram



Class Diagram



Sequence Diagram